



Society of Petroleum Engineers

SPE-229275-MS

Remote Sensing and AI for Monitoring Methane Emissions

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229275-MS](https://doi.org/10.2118/229275-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

Methane emissions are a significant contributor to global warming, and accurate monitoring is crucial for compliance with emission abatement regulations. This paper presents a novel approach to detecting methane emissions using satellite imagery and advanced AI algorithms. The study focuses on some well-studied regions for methane detection, including the Permian Basin, Hassi Messaoud, and Korpezhe. High-resolution imagery acquired from the European Copernicus satellites is employed, and an algorithm inspired by the principles of the multi-band multi-pass (MBMP) method is implemented, focusing on enhancing the distinction between methane-containing pixels and background pixels. The study introduces an ensemble AI framework comprising two complementary models: a machine learning classifier that utilizes textural and spatial feature analysis, and a CNN-based classifier trained on curated image patches. This dual-strategy approach enhances the robustness of the methane detection pipeline by efficiently distinguishing genuine methane signals from artifacts across multiple spectral bands, particularly in B12, B11, and B8a. The results demonstrate the effectiveness of this method, providing detailed spatial and spectral data that enhance the accuracy of emission detection. This research contributes to the energy industry by offering a reliable and scalable method for monitoring methane emissions, supporting regulatory compliance and environmental sustainability efforts.

Introduction

The energy sector is a primary source of methane emissions, a greenhouse gas. Accurate monitoring and quantification of these emissions are crucial for addressing climate change and meeting international regulations aimed at reducing emissions. Effective tracking and measurement strategies in this sector are essential to mitigate the impact of methane on the environment and ensure compliance with global standards. Remote sensing technology, combined with advanced artificial intelligence (AI) and computer vision (CV) algorithms, offers a promising solution for fine detecting methane emissions from satellite imagery. This approach provides a nonintrusive, scalable, and cost-effective method to monitor emissions over large geographic areas. The main objective of this work is to develop a robust solution to fine detect methane emissions using satellite imagery and artificial intelligence algorithms. The scope of the study includes data

collection, preprocessing, algorithm development, and validation. By leveraging high-resolution imagery from the EU's Copernicus satellites (in particular the Copernicus Sentinel-2 mission) we developed a method inspired by the multi-band multi-pass (MBMP) model, aimed at overcoming its limitations in reliably distinguishing methane-containing pixels from background pixels. This research aims to provide accurate and efficient monitoring of methane emissions. The findings will contribute to energy industry emission level acknowledged with a novel and reliable tool, supporting compliance with the Oil and Gas Methane Partnership 2.0 (OGMP 2.0) and other emission abatement regulations. This paper is structured as follows: Section 2 provides a literature review of existing methods for methane emission detection and the role of remote sensing in environmental monitoring. Section 3 details the methods, procedures, and processes involved in data collection, preprocessing, algorithm development, and validation. Section 4 presents the results and observations, highlighting the effectiveness of the proposed solution. Finally, Section 5 concludes the paper with a summary of the key findings and suggestions for future research.

Literature Review

Many recent studies indicate Artificial Intelligence (AI) as an emerging tool to enhance capabilities for environmental monitoring and conservation. Research has been oriented toward wildlife tracking, habitat assessment, and natural disaster prediction by exploiting data acquired from local sensors and processed with deep-learning architectures such as YOLOv4 models [1]. Other works, like [2], show how effective machine learning algorithms such as ANNs and LSTM networks can be as tools for observing surface water quality, where data is collected by in-situ sensors within an IoT infrastructure.

The work presented in [3] provides a relevant review of frameworks for AI techniques in air quality monitoring, emphasizing the importance of state-of-the-art AI models such as convolutional neural networks (CNNs), LSTM networks, transformer architectures, and autoencoders. The study in [4] shows that AI-driven techniques for carbon emission detection and analysis have demonstrated significant potential in enhancing environmental monitoring and sustainability efforts. By leveraging image processing, deep learning models, and data fusion techniques, AI enables accurate and real-time tracking of carbon footprints across urban and industrial settings. However, several challenges persist, including data availability and quality concerns, model interpretability, and ethical and policy considerations.

Regarding methane emission detection, [5] reports that the application of deep learning and computer vision (CNN-based models) has great potential to tackle the important environmental problem of methane emissions and to simplify leak surveys with high accuracy. That study used IR cameras to obtain a large dataset of labelled videos of methane leaks. Many recent articles [6, 7, 8, 9] indicate that satellite images are a good resource for detecting and quantifying methane, in particular exploiting multispectral imagery in the shortwave infrared (SWIR) spectrum in which CH₄ is a strong absorber. For instance, Sentinel5P can provide global-scale observations, but it has relatively low spatial resolution (7 km × 7 km, see Figure 1), so this mission remains inadequate to observe small emissions (< 25 tCH/h) or to attribute emissions to specific facilities in densely equipped oil and gas basins (as indicated in [9]).

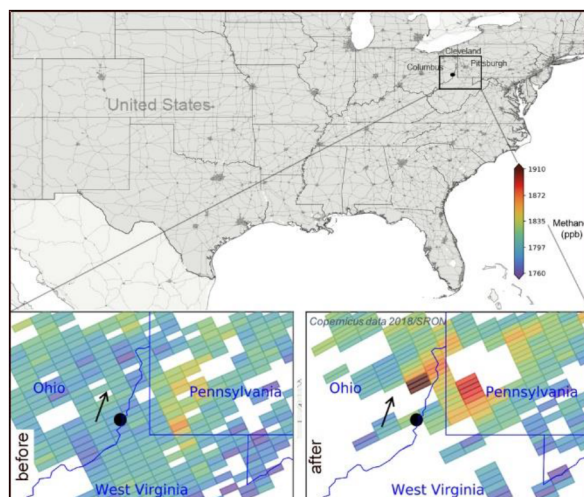


Figure 1—Methane detection example using Sentinel 5P imagery. The spatial resolution of the SWIR images from this satellite is too large to be used in the detection of small methane emissions.

Sentinel2 hyperspectral imagery offers better spatial resolution (up to $10\text{ m} \times 10\text{ m}$ depending on acquired spectral wavelength) and can detect smaller emissions ($< 0.2\text{ tCH/h}$) [10]. As indicated in [9], the accuracy in emission detection from satellites can depend on the number of detected events and sensing conditions. Our goal is to develop a model based on Sentinel2 multispectral imagery (not only SWIR) able to better identify methane plumes even under challenging conditions, for instance when the image background contains many artifacts.

The next section explains the methodology of satellite data analysis performed to obtain the clearest visualization of methane plumes reducing background noise.

Methodology

Input data

This work is based on Copernicus Sentinel2 imagery. This mission consists of two polar orbiting satellites positioned in the same sun-synchronous orbit, with a phase difference of 180° . Sentinel2 aims to monitor changes in land surface conditions. The satellites have a wide swath width (290 km) and a high revisit frequency of 5 days at the Equator, supporting continuous monitoring of changes on the Earth's surface.

The spectral band configuration of the Sentinel2 mission was defined in consultation with the user community during the design phase. Sentinel2 is equipped with an optical instrument payload that samples 13 spectral bands, including:

- four bands at 10 m,
- six bands at 20 m,
- three bands at 60 m spatial resolution.

The orbital swath width is 290 km. The Sentinel2 twin satellites provide image data that largely contribute to ongoing multispectral observations.

Figure 2 reports the spectral bands available from Sentinel-2 observations.

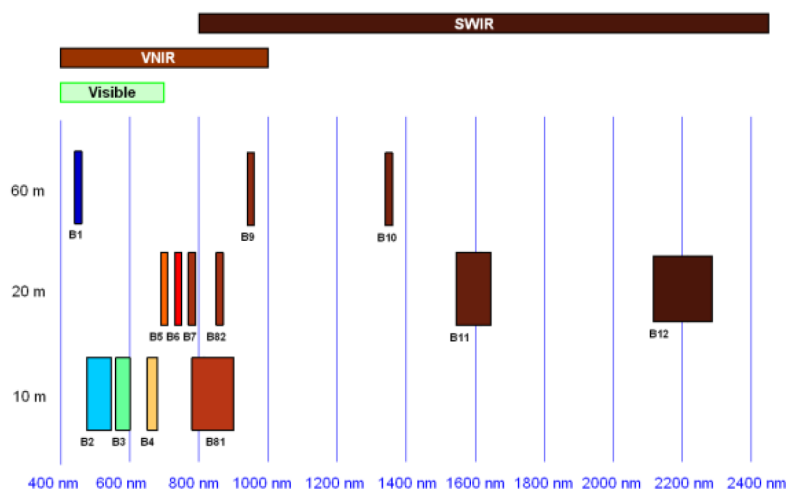


Figure 2—Sentinel-2 spectral bands and resolution. B11 and B12 in SWIR are the bands most relevant on methane absorption, the others can be used to improve the accuracy of the models. This picture is taken from [10]

Data Processing

For our analysis, we focused on the SWIR bands B11 and B12 (20 m × 20 m spatial resolution) to detect the presence of methane. Other bands were used to improve background analysis. For image acquisition, we developed a system to search and download datasets for the location and time interval of interest.

We focused the analysis on specific locations already presented in other papers to enable direct comparison of results ([7, 8, 9]), in particular Hassi Messaoud (Algeria), Korpezhe 1 and Korpezhe 2 (Turkmenistan), and the Permian Basin (USA, TX). Once the images were acquired, preprocessing steps were performed for data normalization on single-band images. The main challenge is distinguishing methane plumes from background and artifacts that can appear in the image for various reasons, mainly due to surfaces that change over time or reflect light differently. The problem is particularly evident in the Permian Basin (USA), where the methane plumes reported in the literature are located near roads and buildings.

To construct a neutral background image of the study area, we applied pixel-wise Huber robust regression using a time series of n historical images of the same location (Cloud Index < 10%). This approach estimates the expected value of each pixel across all historical images for each spectral band, providing robustness against outliers such as transient methane plumes or sensor artifacts. The Huber regression method, introduced by Peter J. Huber [11], offers a compromise between least squares and absolute deviation methods, maintaining efficiency under Gaussian noise while being less sensitive to outliers) [12]. In remote sensing applications, outliers can result from various factors, including surface albedo variations, atmospheric conditions, or instrumental noise. Surface albedo, defined as the fraction of incident light reflected by a surface, can significantly influence the radiance captured by satellite sensors. Variations in albedo, due to changes in land cover or illumination conditions, can introduce anomalies in the data that mimic or obscure actual methane emissions. By employing robust regression techniques, we mitigate the impact of such anomalies, ensuring a more accurate representation of the background.

We utilized the derived regression models to generate a background image for the B12 band (denoted as B12bg) by inferring B12 values from the B11 band. Subsequently, we highlighted methane-related anomalies by computing the difference between the observed B12 image at time t_0 (B12 t_0) and the background image (B12bg).

A similar methodology was applied to images processed using the MBSP (Multi Band Single Pass) [7] method as described by Varon et al. For this, we used the B8a band, which shares the same spatial resolution as the SWIR bands, to perform Huber regression on n prior images. This yielded a background image for the MBSP-processed data (MBSPbg). Methane plumes were then accentuated by subtracting MBSPbg from the MBSP image at time t_0 (MBSP t_0). Each method presents distinct advantages and limitations concerning

methane plume detection and background estimation. To leverage the strengths of both approaches, we computed a weighted sum of the two difference images, with weights inversely proportional to their respective standard deviations. This fusion enhances the detection capability by combining the robustness of each method. Although the method initially required long computation times (due to applying regression to each individual pixel across n images), we optimized the process with GPU-based parallelization using the PyTorch library. As a result, computation times are now reduced to just a few seconds per image.

Validation

To validate the effectiveness of our proposed background estimation and plume enhancement methodology, we conducted a comparative analysis against the Multi Band Multi Pass (MBMP) method on several documented methane release events reported in the literature. Specifically, we selected events from well-characterized sites including Hassi Messaoud (Algeria), Korpezhe 1 and Korpezhe 2 (Turkmenistan), and the Permian Basin (USA, Texas). Our validation focused on statistically quantifying the improvement in separability between pixels containing methane emissions and background pixels without detectable emissions. This assessment was performed using spatially resolved classification metrics, considering both high-concentration zones (typically corresponding to the core of the emission plume) and areas with more diffuse methane presence. We evaluated the discrimination performance by analyzing the distribution of pixel values in the difference images generated by both our method and the MBMP approach. Statistical metrics were employed to measure the effectiveness in enhancing methane signal visibility. The results consistently showed that our method improves the distinction between emission and non-emission pixels, particularly in the plume core, while maintaining or enhancing sensitivity in more spatially diffuse regions.

To minimize the impact of false positives—namely the misclassification of image artifacts as methane-containing pixels—we developed an ensemble artificial intelligence (AI) framework composed of two complementary models: a texture-based machine learning classifier and a convolutional neural network (CNN) classifier. Both models are designed to differentiate genuine methane signals from spurious patterns arising from sensor noise, surface reflectance anomalies, or atmospheric effects.

The first model relies on statistical feature extraction from texture analysis, particularly using the gray-level co-occurrence matrix (GLCM) [13] and related second-order statistical descriptors. GLCM-based features—such as contrast, correlation, energy, and homogeneity—capture spatial relationships between pixel intensities and are effective in characterizing texture irregularities that may arise from artifacts rather than physical emissions. Additional matrices such as the gray-level run length matrix (GLRLM) [14], neighbourhood gray-tone difference matrix (NGTDM) [14], and gray-level size zone matrix (GLSZM) [15] were also considered to improve the model's sensitivity to directional and scale-dependent texture variations.

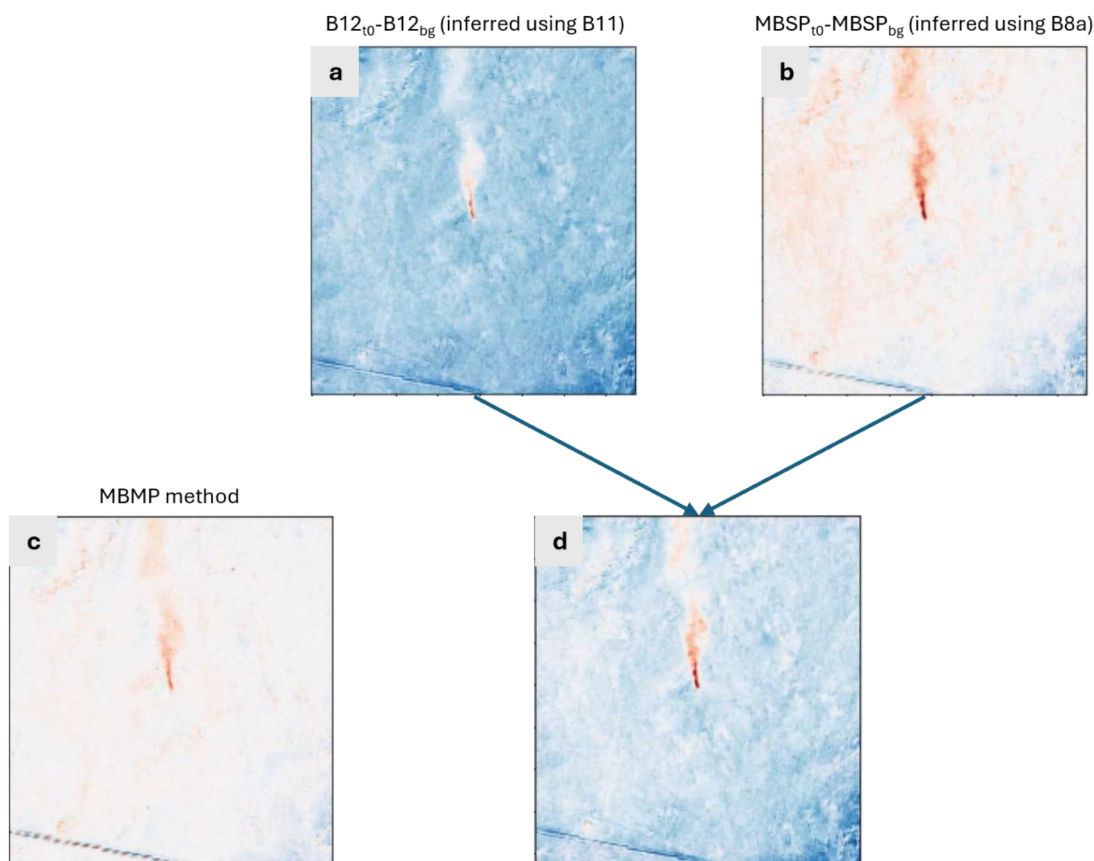


Figure 3—Comparison between the detected plume in the Hassi Messaoud (31.6585° N, 5.9053° E) on 20 November 2019 as reported in [7] and the results obtained with the model developed in this work (see Sect. Results for statistics) (a) B12t0-B12bg retrieval using the last 30 images to calculate background (see Sect. Methodology). (b) The same as panel (a) but for MBSPt0-MBSPbg retrieval. (c) MBMP method [7]. (d) Final result of our method as the weighted sum of the two background retrieval methods

These texture features were computed for each spectral band at the spatial resolution of B12, allowing the model to evaluate whether the spectral anomaly is consistent across bands that are not sensitive to methane absorption. If a signal appears only in B12 and not in other bands such as B11 or B8a, it is more likely to be genuine; conversely, cross-band anomalies may suggest sensor-induced or environmental artifacts. The second component of the ensemble is a CNN-based classifier trained on manually curated image patches that include both true methane plumes and known artifact patterns. The CNN leverages spatial convolutional layers to automatically learn hierarchical features directly from the image data, enabling robust detection of methane-specific spatial signatures while being resilient to background variation. By combining the outputs of these two classifiers through a confidence-weighted scheme, the ensemble model enhances the overall robustness of the methane detection pipeline. This dual-strategy approach allows for both domain-informed feature engineering and data-driven representation learning, increasing accuracy in distinguishing real emissions from confounding artifacts (Fig. 4)

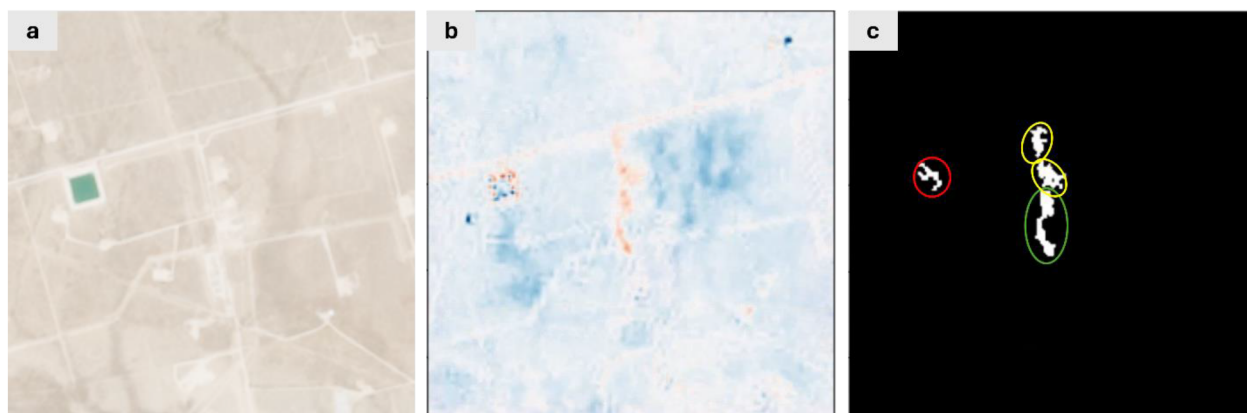


Figure 4—Methane detection result for the Permian Basin release (31.7335, −102.0421) on 11 July 2020. (a) True color image (b) Methane enhancement image generated by our method, showing elevated signal levels associated with the emission event. (c) Classification map of detected high-intensity blobs. The AI-based ensemble model analyzes band B12 along with other bands at matching spatial resolution to distinguish true methane plumes (highlighted in green), potential dispersed methane (highlighted in yellow), and artifacts (highlighted in red), based on spectral and textural consistency across bands.

Results and Observations

The implementation of the developed algorithm on Sentinel2 imagery produced high quality methane emission detection as obtained for the case studies considered. We analysed two representative scenarios to evaluate the performance of our method in enhancing the distinction between methane and background pixels (fig 5): one corresponding to the core of a dense methane plume (fig 5-a), and the other representing a region with spatially dispersed methane presence (fig 5-b). These areas were selected within a known methane release event and are highlighted in the figure, which illustrates the regions of interest used for distributional comparison. Figure 6 presents the estimated probability density functions (PDFs) for methane and background pixels obtained with our method and with the MBMP approach. The probability distributions of methane and background pixel values were estimated using kernel density estimation (KDE), which provides a smooth, continuous approximation of the underlying probability density function. This method offers a more robust and interpretable representation than histograms, particularly for evaluating peak separation and overlap between distributions.

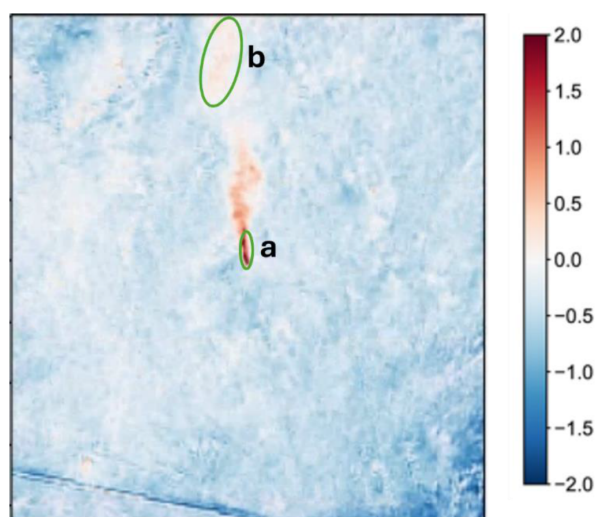


Figure 5—Analyzed areas in Hassi Messaoud (31.6585° N, 5.9053° E) in data 20 November 2019. (a) Center of the methane plume, characterized by high emission intensity. (b) Region with potential presence of spatially dispersed methane. The value range was standardized between −2 and 2 to ensure a consistent comparison with the MBMP image produced using the method by Varon et al.

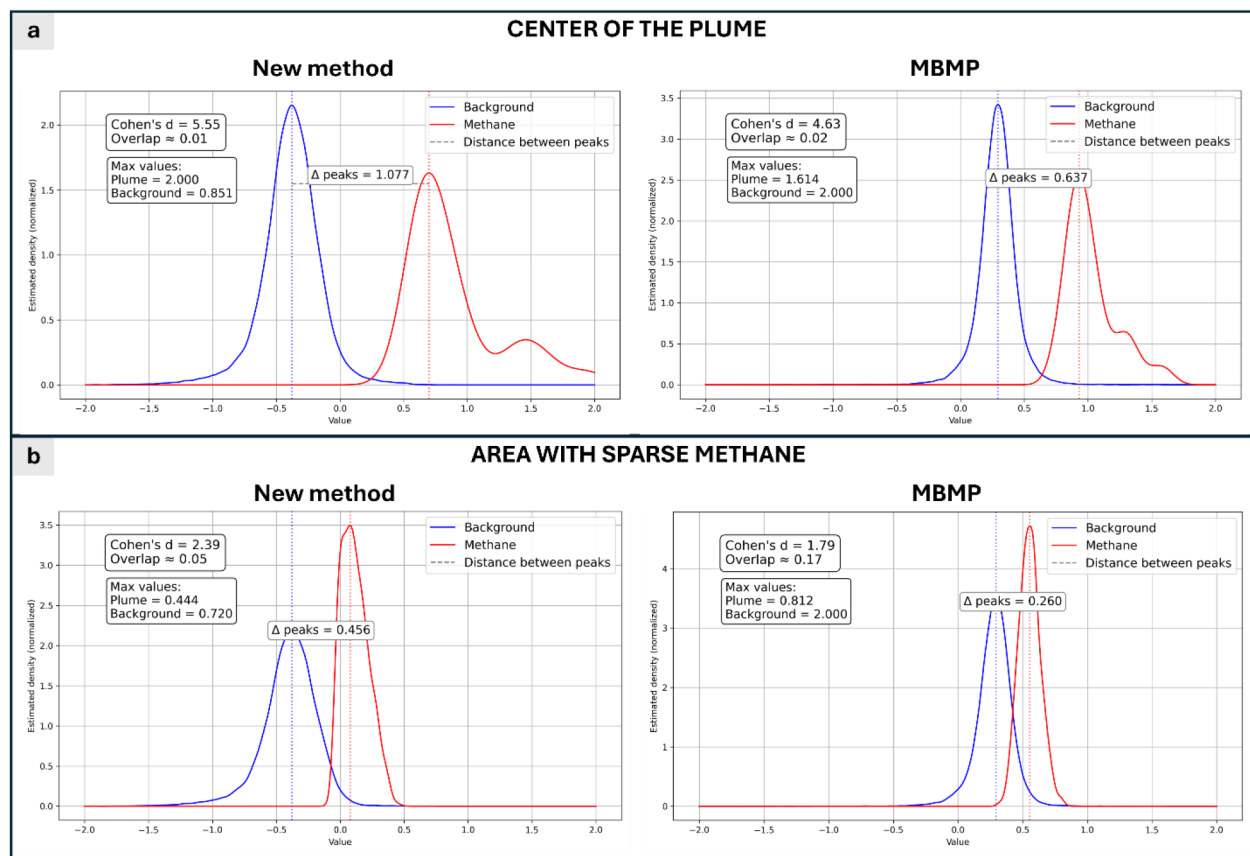


Figure 6—Comparison of background and methane pixel distributions for our method and the MBMP approach. (a) Core of the methane plume. (b) Region with spatially dispersed methane. Each panel shows the estimated probability density functions (KDE) for background and methane pixel values. The figures report the measured distance between distribution peaks, Cohen's d (standardized effect size), the numerical overlap between distributions (area under the minimum of the two density curves), and the maximum value detected in both the methane and background classes. These metrics collectively illustrate the improved separability achieved with our method, particularly in terms of inter-class distinction and spatial accuracy of maximum values.

Figure 6-a focuses on the dense plume core, while Figure 6-b pertains to the region of sparse methane. In both cases, our method produces a greater separation between the peaks of the methane and background distributions. In the dense plume scenario, the inter-peak distance increases from 0.637 with MBMP to 1.077 with our model, representing a 69% relative increase. In the sparse methane region, the distance increases from 0.260 to 0.456, yielding a 75% improvement. These increases suggest an enhanced ability to distinguish emission signals from background noise. To quantify the overlap between the methane and background distributions, we calculated the integral of the pointwise minimum between their respective kernel density estimates (KDEs) over a common domain. This measure represents the total shared area under the two density functions, with values ranging from 0 (no overlap) to 1 (complete overlap), and provides a direct numerical estimate of the degree of distributional similarity.

Despite our method generating a broader distribution for methane pixels compared to MBMP, the numerical overlap between methane and background distributions is reduced in both scenarios, particularly in the case of spatially dispersed methane. This confirms that the increased inter-peak distance is not offset by a loss in separability due to higher variance but rather contributes to a more effective signal-to-background distinction. To further quantify the separation, we computed Cohen's d [16, 17], a standardized effect size metric defined as the difference between the means of the two distributions normalized by their pooled standard deviation. As expected, Cohen's d values were high in both methods, indicating distinct distributions. However, our model consistently produced higher values, reinforcing the conclusion that it offers superior class separability. Moreover, the maximum pixel value detected by our method

is accurately located at the plume center, while in the MBMP result, the highest value is found in a background pixel, and the actual plume core yields lower intensity values. These outcomes confirm that the proposed solution not only replicates established detection patterns but also delivers improvements in clarity, robustness, and computational efficiency (thanks to GPU based parallelization). Beyond the immediate results, the developed algorithm also establishes a valuable foundation for future AI-driven advancements. The high-quality detection outputs generated by this model can be curated as training data for generative AI architectures, for example, diffusion models or generative adversarial networks, designed to synthesize realistic plume patterns [8, 10, 18]. By leveraging such synthetic datasets, it becomes feasible to train detection systems capable of recognizing and quantifying methane plumes with even greater accuracy and adaptability. In operational terms, this opens the door to near real-time monitoring pipelines, where incoming satellite data can be rapidly processed and compared against a generative model's learned representations to flag potential emission events almost instantaneously.

Discussion and Conclusions

The integration of remote sensing and artificial intelligence (AI) technologies represents a significant advancement for the energy industry in its efforts to monitor and mitigate methane emissions, particularly in high-output regions such as the Permian Basin or Hassi Messaoud. This study highlights the effectiveness of combining high-resolution satellite imagery with sophisticated computer vision and machine learning algorithms to detect and monitor methane leaks with a high degree of accuracy and operational efficiency. Central to this approach is the implementation of the new method inspired by multi-band multi-pass (MBMP) algorithm, which enables detailed spatial and spectral analysis of satellite data, allowing for the precise identification of methane plumes and their sources. This methodology not only enhances the reliability of emission estimates but also offers a scalable and adaptable solution that can be applied across various geographic regions and industrial sectors, including agriculture and waste management, which are also significant contributors to global methane emissions. The ability to generate consistent, high-quality data from remote sensing platforms provides regulatory agencies with a powerful tool to enforce emission abatement regulations more effectively, while also enabling energy companies to proactively identify leak sources, implement targeted mitigation strategies, and monitor the success of their interventions over time. In the context of international frameworks such as the Oil and Gas Methane Partnership 2.0 (OGMP 2.0), this research offers a practical and technologically advanced pathway for companies to achieve compliance with stringent emission reduction targets and contribute meaningfully to global climate objectives. Furthermore, the adoption of this approach can lead to substantial operational benefits, including cost savings, improved environmental performance, and enhanced corporate accountability. However, the study also acknowledges certain limitations, such as the potential for interference from other atmospheric gases and aerosols, which can affect detection accuracy.

Addressing these challenges will require the development of more refined algorithms capable of distinguishing methane from other components in the atmosphere. Additionally, the temporal resolution of satellite imagery remains a constraint, as high-frequency monitoring is essential for capturing short-lived or intermittent leaks. Future research should focus on integrating data from multiple satellite platforms and ground-based sensors to improve temporal coverage and provide a more comprehensive understanding of methane emission dynamics. Ultimately, the findings of this study underscore the transformative potential of remote sensing and AI in supporting the energy sector's transition toward a more sustainable and low-carbon future. By enabling accurate, efficient, and scalable methane monitoring, this approach not only facilitates regulatory compliance and environmental stewardship but also empowers stakeholders to make informed decisions that align with long-term climate resilience and sustainability goals.

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